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EXPERIMENT 6

Experiment 6: Time Series Analysis of Website Traffic

AIM:

To analyze the time series data of website traffic to identify trends, seasonality, and anomalies.

DATASET:

A dataset containing daily website traffic data with columns such as Date, Page_Views, Unique_Visitors, and Bounce_Rate.

METHODOLOGY:

1. Load and Clean Data:
o Import the dataset and parse the Date column as a datetime object.
o Handle missing values by interpolation.
2. Time Series Decomposition:
o Decompose the time series data of Page_Views to identify trend, seasonality, and residuals.
3. Visualization:
o Plot the time series data for Page_Views, Unique_Visitors, and Bounce_Rate.
o Use moving averages to smooth the time series and highlight trends.
o Create seasonal plots to visualize patterns over time.
4. Anomaly Detection:
o Identify and visualize anomalies in the time series data using statistical methods such as the Z-score.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose
```

```
# Load dataset
data = pd.read_csv('/content/website_traffic_csv.csv', parse_dates=['Date'],
index_col='Date')

# Time Series Decomposition
decomposition = seasonal_decompose(data['Page_Views'], model='additive')
decomposition.plot()
plt.show()

# Moving Average
data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()

data[['Page_Views', 'Page_Views_MA']].plot()
plt.title('Page Views with Moving Average')
plt.show()

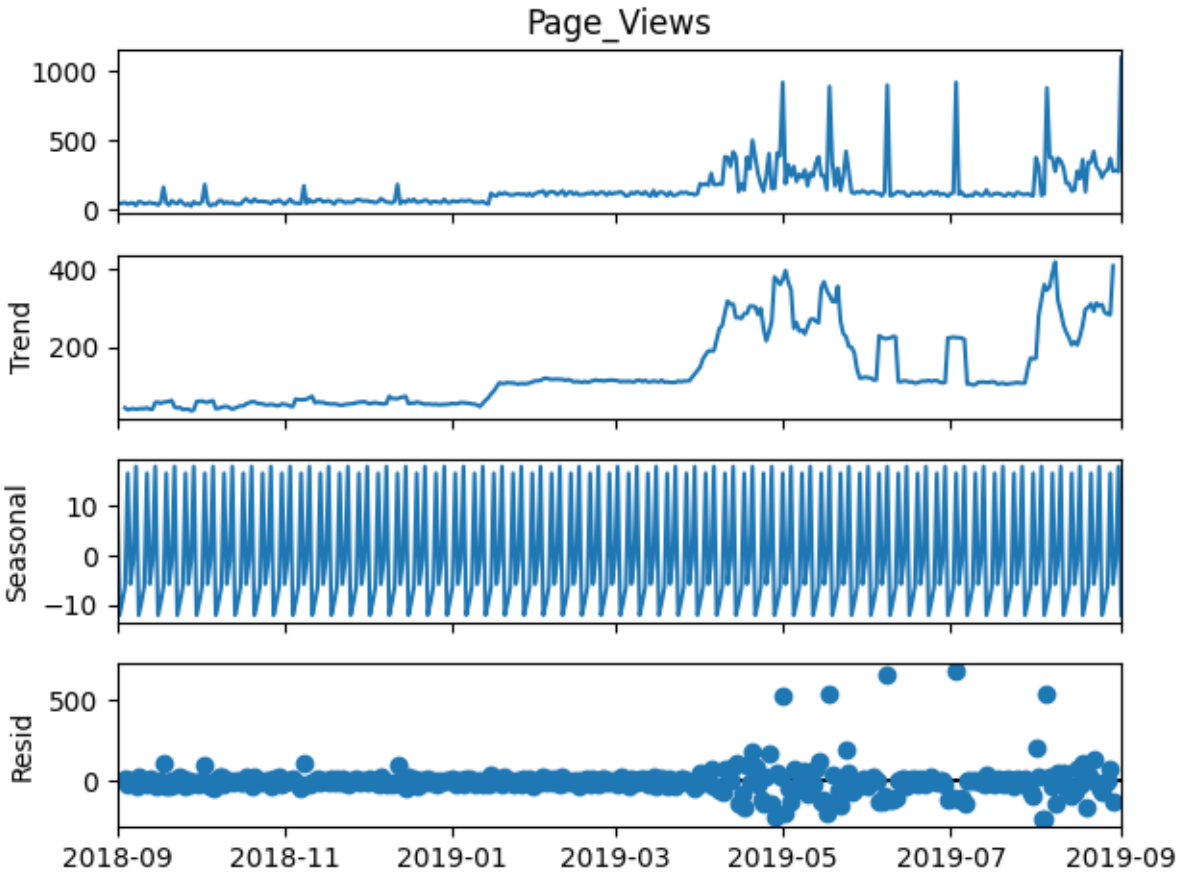
# Anomaly Detection
data['Page_Views_Z'] = (
    data['Page_Views'] - data['Page_Views'].mean()
) / data['Page_Views'].std()

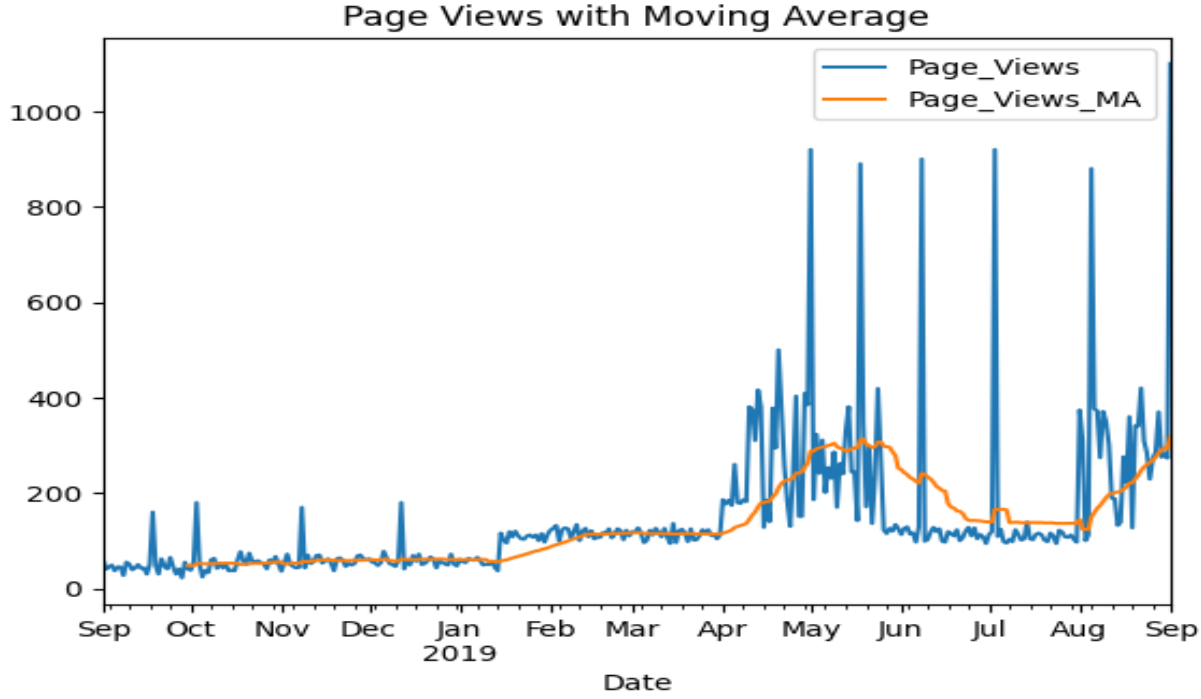
# Find anomalies
anomalies = data[data['Page_Views_Z'].abs() > 3]

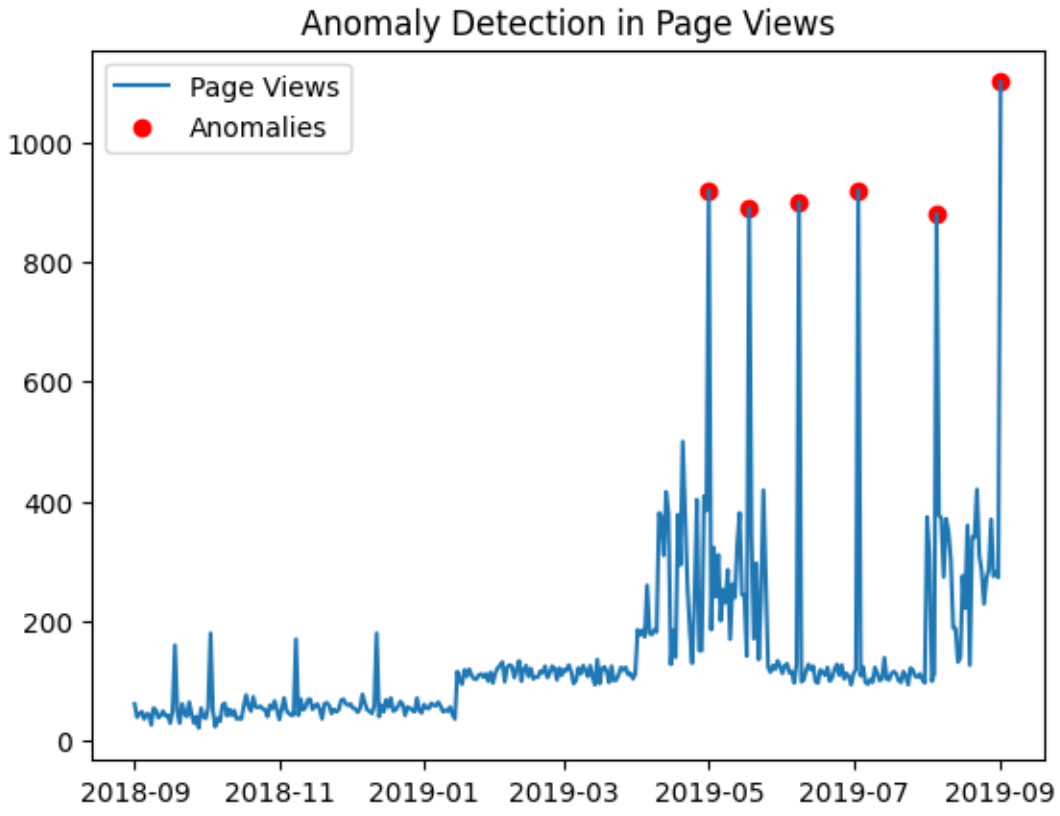
# Plot anomalies
plt.plot(data.index, data['Page_Views'], label='Page Views')
plt.scatter(
    anomalies.index,
    anomalies['Page_Views'],
    color='red',
    label='Anomalies'
)

plt.title('Anomaly Detection in Page Views')
plt.legend()
plt.show()
```

OUTPUT:


Page_Views
1000
500
0
Trend
400
200
Seasonal
10
0
−10
Resid
500
0
2018-09
2018-11
2019-01
2019-03
2019-05
2019-07
2019-09


Page Views with Moving Average
Page_Views
Page_Views_MA
1000
800
600
400
200
0
Sep
Oct
Nov
Dec
Jan
2019
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Date



RESULT:

The website traffic over time is visualized using a line plot, showing the variations and sudden increases in page views.